



Friedrich-Alexander-Universität
Erlangen-Nürnberg

Transport Properties of Charge Carriers in Semiconductors I - Hall Effect in Semiconductors

[ALC - B02]

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Handed in by: Kavan Purohit, Saujanya Dubale

Matriculation number: 23162718, 23148509

Supervisor: Sijia Feng

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Contents

Contents	I
1 Theory	1
1.1 Band Model of Semiconductors	1
1.1.1 Band Structures: Metals, Semiconductors, and Insulators	1
1.1.2 Valence Band, Conduction Band, and Fermi Level	2
1.1.3 Intrinsic and Extrinsic Semiconductors	2
1.1.4 Temperature Dependence of the Fermi Level	3
1.2 Charge Carrier Densities and Electrical Conductivity	4
1.2.1 Intrinsic and Doped Carrier Densities	4
1.2.2 Drude–Sommerfeld Model of Conductivity	4
1.2.3 Scattering Mechanisms and Mobility	5
1.3 Hall Effect	5
1.3.1 Lorentz Force and Hall Voltage	5
1.3.2 Hall Coefficient and Conduction Type	6
1.3.3 Relation to Carrier Density and Mobility	6
1.3.4 Hall Mobility and Hall Scattering Factor	6
1.3.5 One-Band vs. Two-Band Hall Effect	7
2 Experimental Principles	8
2.1 Measurement Protocol Overview	8
2.2 Sample Geometries	8
2.2.1 Silicon Sample (Van der Pauw Geometry)	8
2.2.2 Indium Antimonide Sample (Hall-Bar Geometry)	10
2.3 Measurement of Conductivity	11
2.4 Measurement of the Hall Voltage	12
3 Measurement Setup and Procedure	13
3.1 Setup Overview	13
3.2 Measurement Procedure	15
4 Analysis	18
4.1 Numerical Calculations	18
4.1.1 Band Gap and Fermi Energies	18
4.1.2 Ionisation of Donors and Acceptors	20
4.1.3 Carrier Concentrations	22

4.2	Silicon Sample	23
4.2.1	Temperature-dependent properties	23
4.2.2	Determination of sample parameters	26
4.2.3	Estimate of the Fermi level	26
4.3	Indium Antimonide Sample	27
4.3.1	Temperature-dependent transport properties	27
4.3.2	Validity of the single-band Hall approximation	30
4.3.3	Sample parameters	30
4.3.4	Comparison with literature values	31
4.3.5	Two-band Hall effect	31
	Bibliography	32

1 Theory

1.1 Band Model of Semiconductors

Understanding the electronic transport properties studied in this experiment requires a concise overview of how energy bands form in solids and how these bands determine whether a material behaves as a metal, semiconductor, or insulator. The microscopic basis of this lies in the periodic potential of the crystal lattice, which leads to the formation of allowed energy bands separated by forbidden band gaps [4, 9, 8].

1.1.1 Band Structures: Metals, Semiconductors, and Insulators

The key distinction between conductors, semiconductors, and insulators is the filling of the valence and conduction bands and the size of the band gap E_g . Metals exhibit partially filled bands or overlapping bands and therefore conduct even at low temperatures. Insulators possess a large energy gap, preventing electrons from reaching the conduction band, while semiconductors have a moderate gap that allows thermal excitation of electrons at finite temperature [9].

Figure 1.1 illustrates these differences.

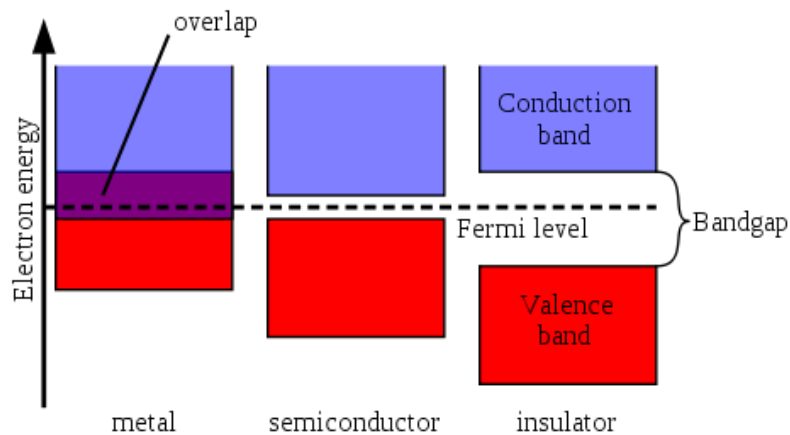


Figure 1.1: Comparison of band gaps in conductors, semiconductors, and insulators [530664]

1.1.2 Valence Band, Conduction Band, and Fermi Level

The *valence band* contains the highest occupied electronic states at $T = 0$, while the *conduction band* contains empty states that can participate in electrical conduction. The *Fermi level* E_F marks the energy at which the occupation probability is $1/2$ at $T = 0$ and governs the distribution of charge carriers at finite temperatures [4].

A schematic band diagram including both doping types is shown in Figure 1.2.

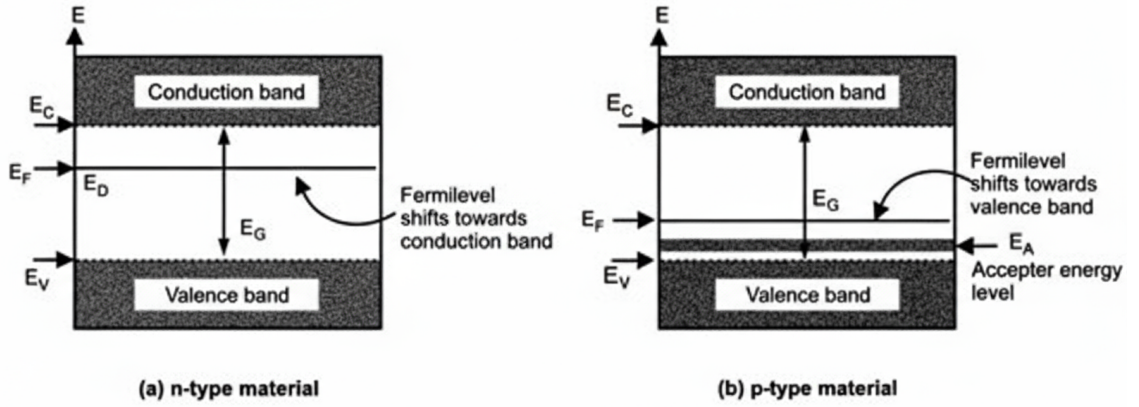


Figure 1.2: Energy band diagram showing valence band, conduction band, Fermi level, and doping levels for n- and p-type materials [9]

1.1.3 Intrinsic and Extrinsic Semiconductors

Intrinsic Semiconductors

An intrinsic semiconductor contains no intentional dopants; all charge carriers originate from thermal excitation of electrons across the band gap. This produces equal concentrations of electrons and holes ($n_i = p_i$). The intrinsic carrier concentration increases exponentially with temperature and is given by

$$n_i(T) = \sqrt{N_C N_V} \exp\left(-\frac{E_g}{2k_B T}\right), \quad (1.1)$$

where N_C and N_V are the effective densities of states in the conduction and valence bands [9, 2].

Extrinsic Semiconductors: Donors and Acceptors

Doping introduces additional energy levels near the band edges. Donor atoms (e.g. P, As) provide weakly bound electrons, shifting the Fermi level towards the conduction band, whereas acceptor atoms (e.g. B, Al) create holes and shift the Fermi level towards the valence band [9].

When both donors and acceptors are present, partial neutralization occurs. This *compensation* modifies the effective charge carrier concentration and must be considered when solving the neutrality equation in temperature-dependent transport analysis.

1.1.4 Temperature Dependence of the Fermi Level

The Fermi level $E_F(T)$ is obtained from the *charge neutrality equation*, which balances mobile carriers with ionized dopants:

$$n(T, E_F) + N_A^-(T, E_F) = p(T, E_F) + N_D^+(T, E_F). \quad (1.2)$$

This equation determines how E_F shifts with temperature in intrinsic and doped materials. Its solution is essential for interpreting temperature-dependent carrier densities in Hall measurements [2, 8].

Figure 1.3 shows a typical temperature dependence of the carrier concentration, including freeze-out, extrinsic, and intrinsic regimes.

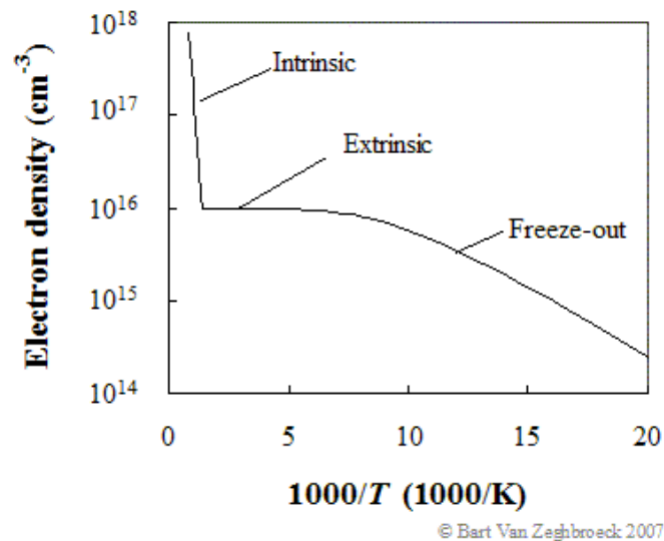


Figure 1.3: Typical carrier concentration in a doped semiconductor as a function of inverse temperature, showing freeze-out, extrinsic, and intrinsic regions [10]

1.2 Charge Carrier Densities and Electrical Conductivity

Electrical transport in semiconductors is governed by the concentration of charge carriers and by their mobility. Both quantities depend strongly on temperature and material parameters such as the band gap, effective density of states, and dopant concentrations. In this section we summarize the relations relevant for the later Hall-effect analysis, following [9, 8, 2, 4].

1.2.1 Intrinsic and Doped Carrier Densities

At thermal equilibrium, the electron concentration n and hole concentration p in a semiconductor are determined by the effective densities of states N_C and N_V and by the position of the Fermi level E_F .

For an intrinsic semiconductor, electrons and holes are thermally generated in pairs, yielding

$$n_i = p_i = \sqrt{N_C N_V} \exp\left(-\frac{E_g}{2k_B T}\right), \quad (1.3)$$

where E_g is the band gap. This exponential dependence produces the characteristic steep rise of intrinsic carrier concentration with temperature [9].

In doped semiconductors, ionized donors and acceptors modify the free carrier density. Charge neutrality requires

$$n + N_A^- = p + N_D^+, \quad (1.4)$$

where N_D^+ and N_A^- are the concentrations of ionized donors and acceptors, respectively. Determining $E_F(T)$ from Eq. (1.4) allows computation of the temperature-dependent carrier densities used in the evaluation of the experiment [2].

A schematic example of this characteristic temperature regimes in a doped semiconductor is already shown in Figure 1.3.

1.2.2 Drude–Sommerfeld Model of Conductivity

Charge carrier transport under an applied electric field is described by the Drude–Sommerfeld model [4]. The average drift velocity v_d of a charge carrier is proportional to the electric field E ,

$$v_d = \mu E, \quad (1.5)$$

where μ is the carrier mobility.

The electrical conductivity follows as

$$\sigma = q (n \mu_e + p \mu_h), \quad (1.6)$$

with q the elementary charge and μ_e, μ_h the electron and hole mobilities. When one carrier type dominates,

$$\sigma = qn\mu_e \quad (\text{n-type}), \quad \sigma = qp\mu_h \quad (\text{p-type}). \quad (1.7)$$

1.2.3 Scattering Mechanisms and Mobility

Carrier mobility is limited by scattering. Two mechanisms dominate [9, 8]:

- **Ionized impurity scattering** (low T): mobility increases with temperature.
- **Acoustic phonon scattering** (high T): mobility decreases approximately as

$$\mu_{ac} \propto T^{-3/2}. \quad (1.8)$$

These temperature dependencies are essential for interpreting the Hall mobility and conductivity curves measured in the experiment.

1.3 Hall Effect

The Hall effect provides an essential method for determining the type and density of charge carriers in semiconductors. When combined with conductivity measurements, it allows extraction of the Hall mobility and enables distinction between one-band and multi-band transport regimes [9, 8, 2].

1.3.1 Lorentz Force and Hall Voltage

When a magnetic field $\mathbf{B}$ is applied perpendicular to a current density $\mathbf{J}$ in a semiconductor, charge carriers experience the Lorentz force,

$$\mathbf{F}_L = q (\mathbf{v} \times \mathbf{B}), \quad (1.9)$$

which deflects electrons or holes toward one side of the sample. This leads to a charge imbalance and the buildup of a transverse electric field E_H , producing the Hall voltage V_H ,

$$V_H = E_H w, \quad (1.10)$$

where w is the sample width. In steady state, the Hall field balances the Lorentz force, yielding the basis for determining the Hall coefficient.

1.3.2 Hall Coefficient and Conduction Type

The Hall coefficient is defined through the relation

$$E_H = R_H J B, \quad (1.11)$$

and for a semiconductor with a single dominant carrier type (one-band approximation),

$$R_H = \frac{1}{qn} \quad (\text{electrons}), \quad R_H = -\frac{1}{qp} \quad (\text{holes}). \quad (1.12)$$

Thus, the *sign* of R_H reveals the conduction type: positive for holes (p-type), negative for electrons (n-type). The magnitude determines the majority carrier concentration.

1.3.3 Relation to Carrier Density and Mobility

Combining the Hall effect with the Drude conductivity expression

$$\sigma = q(n\mu_e + p\mu_h), \quad (1.13)$$

allows independent determination of the mobility. For a one-band conductor,

$$\mu = \frac{\sigma}{qn} = \frac{\sigma |R_H|}{1}. \quad (1.14)$$

Thus the Hall effect resolves the ambiguity present in conductivity measurements, where only the product $n\mu$ or $p\mu$ appears.

1.3.4 Hall Mobility and Hall Scattering Factor

The Hall mobility is defined as

$$\mu_H = |R_H| \sigma. \quad (1.15)$$

It differs from the drift mobility μ by the Hall scattering factor r_H [8],

$$\mu_H = r_H \mu. \quad (1.16)$$

The value of r_H depends on the dominant scattering mechanism and typically lies between 1 and 2. In this experiment, $r_H = 1$ is assumed, consistent with the approximation used in standard semiconductor transport analysis [2].

1.3.5 One-Band vs. Two-Band Hall Effect

The one-band expressions above are valid when either electrons or holes dominate conduction. However, in materials such as InSb, both carrier types may contribute significantly at certain temperatures. In such a two-band scenario, the Hall coefficient becomes

$$R_H = \frac{p \mu_h^2 - n \mu_e^2}{q (p \mu_h + n \mu_e)^2}, \quad (1.17)$$

which may change sign with temperature. This phenomenon is especially relevant for the InSb sample in the second part of the experiment, where mixed conduction occurs in an intermediate temperature range [9].

2 Experimental Principles

This experiment investigates the temperature-dependent transport parameters of two semiconductor samples: silicon (Si) in van-der-Pauw geometry and indium antimonide (InSb) in Hall-bar geometry. For both samples, conductivity, Hall coefficient, carrier density and Hall mobility are extracted using full I–V characteristics at each contact configuration.

2.1 Measurement Protocol Overview

A source-measure unit (SMU) provides the current, while a second SMU records the voltage through a switch matrix that selects the required contact configuration. All measurements are performed in the four-terminal configuration, and the applied voltage is kept below 0.5 V to suppress surface currents and undesired heating effects [3].

Figure ?? shows the two samples used in the experiment— indium antimonide (InSb) in Hall-bar geometry and silicon (Si) in van-der-Pauw geometry.

2.2 Sample Geometries

2.2.1 Silicon Sample (Van der Pauw Geometry)

The silicon sample is a thin, square lamella with four contacts located along its perimeter. This arrangement satisfies the van-der-Pauw requirements of uniform thickness, isotropic material

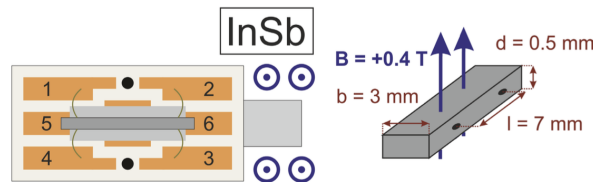


Figure 2.1: (a) The InSb sample arranged in a Hall-bar geometry, showing the contact positions and the direction of the applied magnetic field.

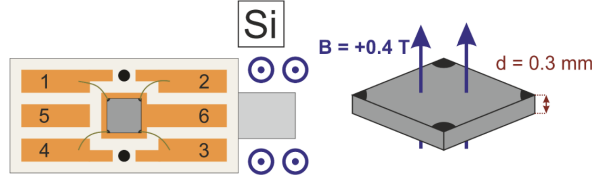


Figure 2.2: (b) The Si sample in Van der Pauw geometry, with marked contact points and the orientation of the magnetic field indicated.

properties, and point-like boundary contacts, allowing the sheet resistance to be determined independent of the exact sample shape.

To obtain the resistivity, two characteristic configurations are measured:

$$R_{23,41} = \frac{V_{23,41}}{I}, \quad R_{34,12} = \frac{V_{34,12}}{I},$$

corresponding to two orthogonal current–voltage paths. Any asymmetry between these configurations is accounted for by the correction factor

$$x = \frac{R_{23,41}}{R_{12,34}}.$$

The resistivity is then given by

$$\rho = \frac{\pi t}{\ln 2} \frac{V_{23,41} + V_{34,12}}{2I} f(x),$$

from which the conductivity follows.

Hall measurement. When a magnetic field B is applied perpendicular to the sample plane, the Hall voltage is determined using the reversal scheme specified in the experiment instructions. For the van-der-Pauw configuration, the Hall voltage is evaluated using the logarithmic expression

$$V_H = \frac{2I}{d} \ln \left(\frac{V_{23,41} + V_{34,12}}{V_{12,34} + V_{24,13}} \right),$$

where d is the sample thickness.

The Hall coefficient follows directly as

$$R_H = \frac{V_H}{B}.$$

Its sign indicates the conduction type of the Si sample. Once R_H and ρ are known, the carrier

density and Hall mobility are obtained through

$$n = \frac{1}{|eR_H|}, \quad \mu_H = |R_H| \sigma.$$

2.2.2 Indium Antimonide Sample (Hall-Bar Geometry)

The indium antimonide (InSb) sample is fabricated in a conventional Hall-bar layout, a geometry designed to establish a well-defined longitudinal current path while providing separate transverse contacts for Hall-voltage detection [2]. The bar consists of a narrow conducting channel of width b and length l , with additional contact pairs placed symmetrically along its sides. Such a layout is particularly suited for high-mobility semiconductors like InSb, where even small geometrical asymmetries can influence the measured voltages.

Longitudinal (conductivity) measurement. The longitudinal resistance of the Hall bar is obtained by driving current through contacts 5–6 while measuring the corresponding voltage between contacts 1–2. To account for residual geometrical asymmetries, a second measurement is taken with the voltage probes interchanged, yielding two values,

$$V_{56,12}, \quad V_{65,12}.$$

Using their average, the resistivity of the rectangular bar becomes

$$\rho = \frac{bd}{l} \frac{V_{56,12} + V_{65,12}}{2I},$$

from which the conductivity follows as $\sigma = 1/\rho$. Here, d is the sample thickness, and the geometric prefactor $l/(bd)$ converts the measured resistance into a material property according to Ohm's law.

Hall-voltage measurement. For the Hall measurement, the same current path (5–6) is used, while the transverse voltage is recorded between contacts 1 and 4. Two readings,

$$V_{56,14}, \quad V_{65,14},$$

are again taken to minimise contributions from unavoidable contact misalignment. The Hall coefficient for a rectangular Hall bar is obtained from

$$R_H = \frac{d}{IB} \frac{V_{56,14} + V_{65,14}}{2},$$

where B is the applied magnetic flux density.

Physical relevance for InSb. InSb has an exceptionally small band gap ($E_g \approx 0.17\text{--}0.23\text{ eV}$, depending on temperature [1]) and very high carrier mobility. Consequently, the material can exhibit a transition from extrinsic to intrinsic conduction over the temperature range of the experiment. Accurate determination of the Hall voltage is therefore essential, as the sign and magnitude of R_H directly indicate which carrier type dominates. Finally, combining R_H with the measured conductivity yields the Hall mobility,

$$\mu_H = |R_H| \sigma,$$

which provides insight into the dominant scattering mechanisms in InSb.

2.3 Measurement of Conductivity

The conductivity of both semiconductor samples is determined from their longitudinal resistance using a four-terminal configuration, ensuring that only the intrinsic voltage drop across the material is measured [2, 8]. During each measurement, the source voltage is swept between -0.5 V and $+0.5\text{ V}$, and the resulting I - V curve is used to extract the differential resistance from its slope,

$$R = \frac{dV}{dI}.$$

Operating within this low-voltage window avoids heating effects and suppresses unwanted surface currents, which is especially important for high-mobility materials.

To eliminate offset voltages arising from thermoelectric effects or instrumental drift, the current is applied in both polarities and the symmetric voltage component is evaluated as

$$V_{\text{sym}} = \frac{V(+I) - V(-I)}{2}.$$

This provides a reliable measure of the resistive contribution across the full temperature range.

The resistivity ρ is then obtained according to the geometry of the respective sample: for the silicon sample, the van-der-Pauw relations are used, whereas for the InSb Hall bar, ρ follows directly from the dimensions of the rectangular bar through Ohm's law. The conductivity is finally obtained via

$$\sigma = \rho^{-1},$$

and serves as one of the key quantities combined with Hall-effect data to determine carrier density and mobility.

2.4 Measurement of the Hall Voltage

The Hall voltage is measured by applying a current through the sample while a magnetic field is directed perpendicular to its surface. In the presence of a single dominant carrier type, the Lorentz force creates a transverse electric field that develops into a measurable Hall voltage, which is directly related to the carrier density [2].

To ensure that only the Hall contribution is retained, the measurement is performed for both current polarities and both magnetic-field directions. Combining these four configurations suppresses offsets arising from contact misalignment, thermoelectric effects, and longitudinal voltage leakage [8]. The resulting Hall voltage is the antisymmetric component,

$$V_H = \frac{V(+B, +I) - V(+B, -I) - V(-B, +I) + V(-B, -I)}{4}.$$

Hall coefficient and carrier density. With the sample thickness d and applied field B , the Hall coefficient follows as

$$R_H = \frac{V_H d}{IB}.$$

For a one-band semiconductor, the carrier density is then

$$n = \frac{1}{|eR_H|},$$

and the sign of R_H identifies whether electrons or holes dominate the conduction.

Hall mobility. Combining the Hall coefficient with the measured conductivity yields the Hall mobility,

$$\mu_H = |R_H| \sigma,$$

which reflects how efficiently carriers respond to an applied electric field and provides insight into the strength of scattering mechanisms across the temperature range [8, 1].

Applicability to Si and InSb. For the silicon sample, the one-band description remains valid over the full temperature range. In contrast, the narrow band gap of InSb ($E_g \approx 0.17\text{--}0.23$ eV [1]) leads to significant intrinsic carrier generation at moderate temperatures. When both electrons and holes contribute, the simple relation $n = 1/(eR_H)$ no longer holds and the Hall coefficient can even change sign, indicating a transition into a two-band conduction regime.

3 Measurement Setup and Procedure

The Hall-effect measurements carried out in this work rely on a semiautomated setup that combines precise temperature control, flexible electrical switching, and reproducible magnetic-field operation. This section provides a concise overview of the apparatus and outlines the experimental procedure used to acquire the temperature-dependent transport parameters. The presentation follows the general structure of modern semiconductor characterization experiments, as described in standard literature such as [4, 8, 2, 9] and in the laboratory guidelines [3].

3.1 Setup Overview

The experiment is carried out inside a closed cryogenic environment that provides a stable and reproducible thermal setting for Hall and conductivity measurements over a wide temperature range. The essential components of the setup and their roles are summarised below. For clarity, the overall wiring and device layout closely follows the schematic provided in the laboratory instructions [3]. This is shown in Figure 3.1, at the end of this section.

In combination, these devices form a fully automated measurement chain: the SMUs supply the excitation current and record the corresponding voltages, the switch unit selects the appropriate contact configuration, the electromagnet applies the required transverse field for Hall measurements, and the cryostat with the PT1000 sensor maintains the thermal environment. All components are synchronised via the NOMAD CAMELS software [3], which coordinates the measurement sequence, contact switching, magnetic-field polarity changes, and temperature stabilisation. This integration minimises manual intervention and ensures reproducible acquisition of temperature-dependent transport parameters.

- **Cryostat and Temperature Control:** The semiconductor sample is mounted inside a vacuum cryostat equipped with a heater and a liquid-nitrogen cooling line. A PID controller regulates the temperature during the measurement, enabling cooling down to approximately 100 K and maintaining thermal stability at each setpoint. The system corresponds to the cryostat described in the instruction manual [3].
- **Source-Measure Units (SMUs):** Two high-precision SMUs provide the drive current and measure the resulting voltages. Their functions match the units listed in the hardware

appendix of the instructions [3]. One SMU sources the excitation current, while the second measures differential voltages under various geometries. This setup enables both four-terminal resistivity measurements and Hall-voltage measurements under magnetic-field reversal.

- **Switch Unit for Contact Selection:** A computer-controlled relay matrix (Agilent 34972A with 34904A switch card) automatically routes SMU terminals to the appropriate sample contacts. This avoids manual rewiring and allows rapid cycling through the van der Pauw and Hall-bar configurations. The relay layout is documented in the laboratory instructions [3].
- **Electromagnet and Polarity Control:** A transverse magnetic field is provided by an electromagnet that can reverse its polarity via a programmable control unit. As recommended in [3], field reversal is carried out during each Hall measurement to suppress systematic offsets. The magnet control incorporates a soft-ramp procedure to avoid inductive voltage spikes.
- **Temperature Sensor (PT1000):** A four-wire PT1000 resistor placed in close proximity to the sample monitors the actual temperature. The resistance is read by the multimeter module of the switch unit (34901A), which internally converts it to temperature. This ensures precise feedback for the PID controller and stable conditions during the temperature scan.

Notes on Calibration and Noise Reduction. To maintain consistent measurement conditions, the SMUs are allowed to thermally stabilise before beginning the protocol, and the magnet controller uses a voltage ramp to avoid inductive transients. Between measurements, the switch-unit relays are opened to prevent unintended current paths, following the recommended safety sequence in the manual [3]. Hall-voltage offsets are suppressed by averaging measurements taken under positive and negative magnetic-field polarities. The PT1000 is continuously logged to ensure the PID loop maintains its stability window, particularly during cryogenic operation where temperature drifts can influence resistance measurements. These steps collectively reduce systematic errors and improve the accuracy of the extracted transport parameters.

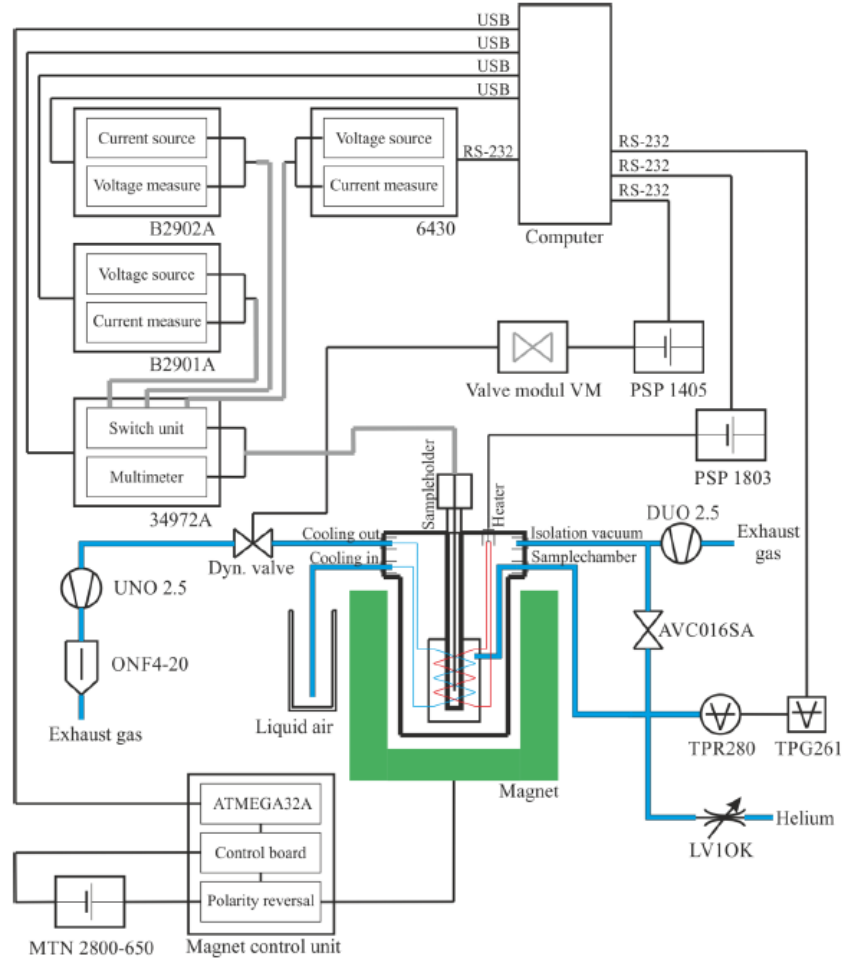


Figure 3.1: Schematic overview of the measurement setup showing the cryostat, SMUs, switch unit, temperature controller and electromagnet. Adapted using information from the official experiment instructions [3].

3.2 Measurement Procedure

The measurement sequence follows a structured protocol to ensure well-defined electrical contacts, stable thermal conditions, and reproducible transport data across the full temperature range. The overall logic of the procedure is consistent with the workflow described in the laboratory manual [3]. Suitable figure placements are indicated below, especially for contact geometries and temperature-scan schemes.

1. **Verification of Electrical Contacts:** Before initiating any temperature-dependent measurements, I–V characteristics are acquired for all relevant contact pairs. A linear response

with minimal series resistance confirms correct wiring and reliable four-terminal measurement conditions. This step corresponds to the initial *I–V characteristics* check recommended in the instructions [3].

2. **Cooling and Temperature Stabilisation:** The sample is cooled to the lowest target temperature (typically around 100 K). Once reached, the PID controller is activated to stabilise the temperature within a narrow tolerance. Only after achieving thermal equilibrium is the measurement protocol continued.
3. **Execution of Automated Measurement Protocol:** A predefined protocol coordinates the switch unit, SMUs, and the electromagnet using the NOMAD CAMELS framework [3].
 - Conductivity is obtained from I–V sweeps under the appropriate contact configuration.
 - Hall voltages are recorded for both positive and negative magnetic-field polarities, as required to eliminate spurious offsets.
 - The resulting slopes from the voltage–current fits allow extraction of sheet resistance, Hall coefficient, and Hall mobility according to standard semiconductor characterisation methods [8, 7].
4. **Temperature-Dependent Measurement Loop:** A looped routine increases the temperature in discrete steps. For each temperature T :
 - the PID controller stabilises the setpoint,
 - the full conductivity and Hall-measurement sequence is executed,
 - and the measured temperature is stored together with the electrical quantities.

This follows the recommended *Stabilize PID + Run Subprotocol* procedure outlined in the manual [3].

5. **Recorded Quantities:** For each stabilised temperature point, the resulting transport quantities are:

$$\sigma(T), \quad R_H(T), \quad n(T) = \frac{1}{e|R_H|}, \quad \mu_H(T) = |R_H|\sigma(T).$$

These parameters enable determination of conduction type, doping concentration, activation energies, and scattering mechanisms, aligning with typical semiconductor analysis workflows [2, 4, 9].

Suppressing Systematic Errors. To ensure accurate Hall-voltage extraction, each measurement is performed for both magnetic-field polarities and averaged to remove background offsets. Between individual measurement steps, the relay matrix is opened to avoid unintended current paths, as recommended in the protocol sequence [3]. The magnet controller employs a soft-ramp procedure to avoid inductive spikes, ensuring well-defined field conditions.

Data Handling and Logging. All measurement variables—including voltages, currents, PID temperature readings, and fit results—are automatically logged by the NOMAD CAMELS environment. This allows reproducible post-processing and simplifies the generation of the temperature-dependent plots required for the analysis (conductivity, Hall coefficient, mobility, and charge-carrier density). Temperature readings from the PT1000 are stored alongside each electrical measurement point to verify thermal stability throughout the scan.

4 Analysis

4.1 Numerical Calculations

In order to analyse the temperature dependence of semiconductor properties, we numerically implemented the expressions for the band gap energy, intrinsic Fermi energy, effective densities of states, ionised dopant concentrations and charge carrier concentrations. All computations were performed in Python using the full temperature range $T \in [20, 1000]$ K. The relevant material parameters follow the experiment instructions, with the valence-band edge $E_v = 0$, the conduction-band edge $E_c = E_g(T)$, and the donor and acceptor ionisation energies $\Delta E_D = \Delta E_A = 0.045$ eV.

4.1.1 Band Gap and Fermi Energies

The temperature-dependent band gap was modelled using

$$E_g(T) = 1.17 \text{ eV} - 4.73 \times 10^{-4} \text{ eV/K} \frac{T^2}{T + 636}, \quad (4.1)$$

which decreases monotonically with increasing temperature due to lattice expansion and electron-phonon interactions.

The effective densities of states in the conduction and valence bands are given by

$$N_c(T) = 2M_c \left(\frac{2\pi m_{e,\text{eff}} k_B T}{h^2} \right)^{3/2}, \quad (4.2)$$

$$N_v(T) = 2M_v \left(\frac{2\pi m_{h,\text{eff}} k_B T}{h^2} \right)^{3/2}, \quad (4.3)$$

with effective masses $m_{e,\text{eff}} = 0.36 m_e$, $m_{h,\text{eff}} = 0.81 m_e$, and valley degeneracies $M_c = 6$, $M_v = 1$.

The intrinsic Fermi level is then

$$E_i(T) = \frac{E_g(T)}{2} + \frac{k_B T}{2} \ln \left(\frac{N_v(T)}{N_c(T)} \right). \quad (4.4)$$

To obtain the extrinsic Fermi level $E_F(T)$ for doped silicon, we solve the neutrality condition iteratively:

$$n(T, E_F) + N_A^-(T, E_F) = p(T, E_F) + N_D^+(T, E_F), \quad (4.5)$$

where

$$n(T, E_F) = N_c(T) \exp\left[-\frac{E_c - E_F}{k_B T}\right], \quad (4.6)$$

$$p(T, E_F) = N_v(T) \exp\left[-\frac{E_F - E_v}{k_B T}\right]. \quad (4.7)$$

The resulting temperature dependence of $E_g(T)$, $E_i(T)$, $E_F(T)$ and the donor/acceptor energy levels is shown in Fig. 4.1.

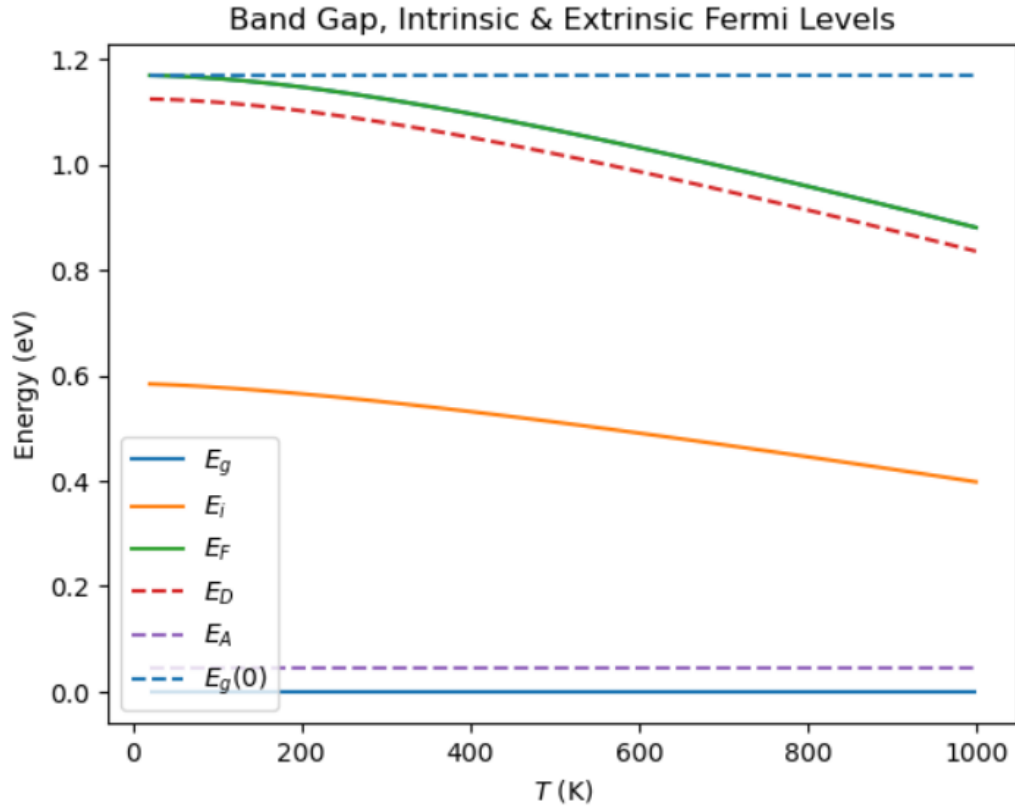


Figure 4.1: Temperature dependence of the band gap energy $E_g(T)$, intrinsic Fermi level $E_i(T)$, extrinsic Fermi level $E_F(T)$, and donor/acceptor energy levels using the numerical model.

4.1.2 Ionisation of Donors and Acceptors

The fraction of ionised donors and acceptors was computed using

$$N_D^+(T, E_F) = \frac{N_D}{1 + g_D \exp\left(\frac{\Delta E_D - E_F}{k_B T}\right)}, \quad (4.8)$$

$$N_A^-(T, E_F) = \frac{N_A}{1 + g_A \exp\left(\frac{E_F - \Delta E_A}{k_B T}\right)}, \quad (4.9)$$

with degeneracies $g_D = 2$ and $g_A = 4$, and doping concentrations $N_D = 10^{16} \text{ cm}^{-3}$, $N_A = 10^{13} \text{ cm}^{-3}$.

Figure 4.2 shows that donors are almost fully ionised for the entire temperature range, whereas acceptors remain only weakly ionised and never approach their nominal concentration. This behaviour is consistent with the strong donor dominance in the sample.

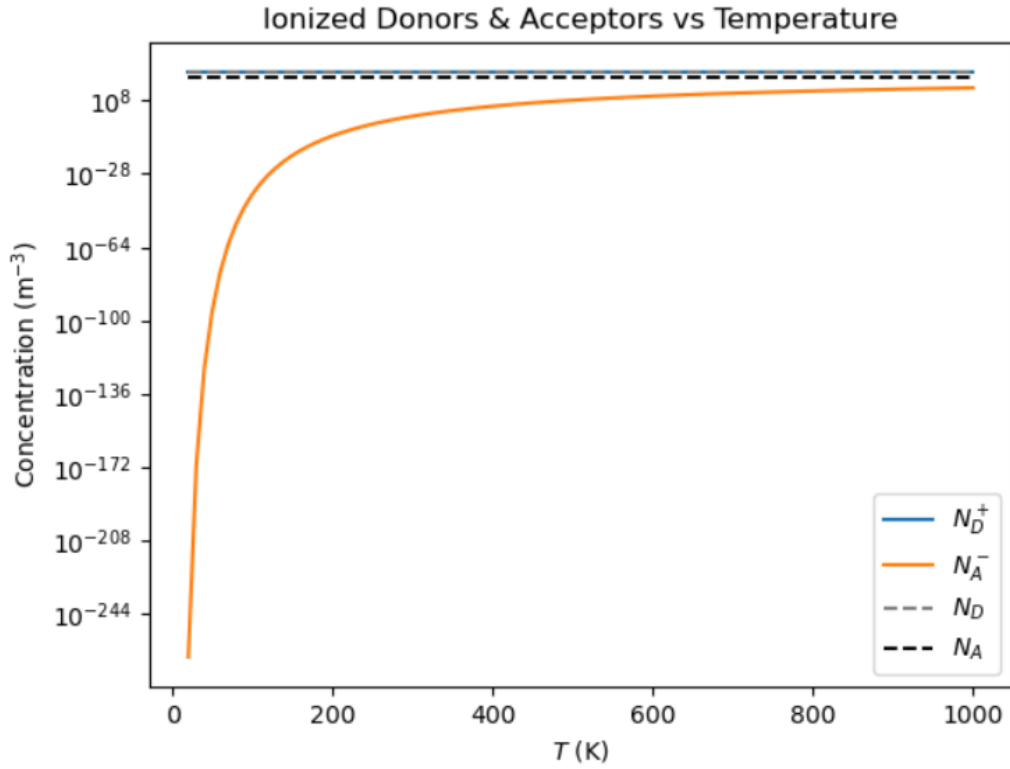


Figure 4.2: Concentration of ionised donors N_D^+ and acceptors N_A^- as a function of temperature. Donors are strongly ionised, whereas acceptors remain far below their nominal concentration.

Because $N_A^-(T)$ is orders of magnitude smaller than N_A , a zoom of the region $0.9N_A \leq N_A^- \leq 1.1N_A$ shows only the horizontal reference line (Fig. 4.3). A second zoom, adjusted to the actual magnitude of $N_A^-(T)$, is included in Fig. 4.4.

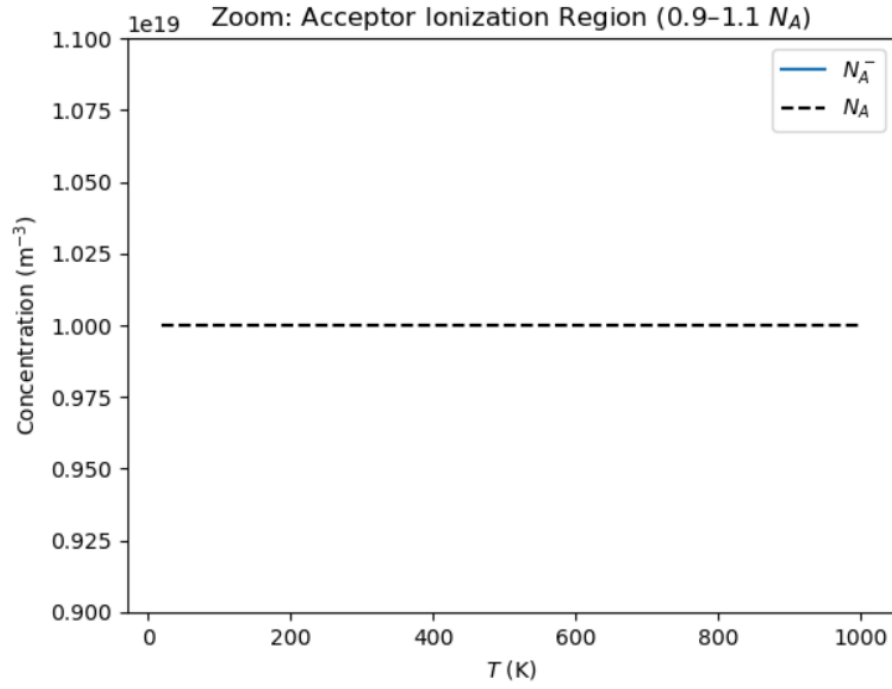


Figure 4.3: Zoom of the region $0.9N_A$ – $1.1N_A$. Since $N_A^-(T)$ never approaches N_A , the ionised acceptor curve lies completely outside this range.

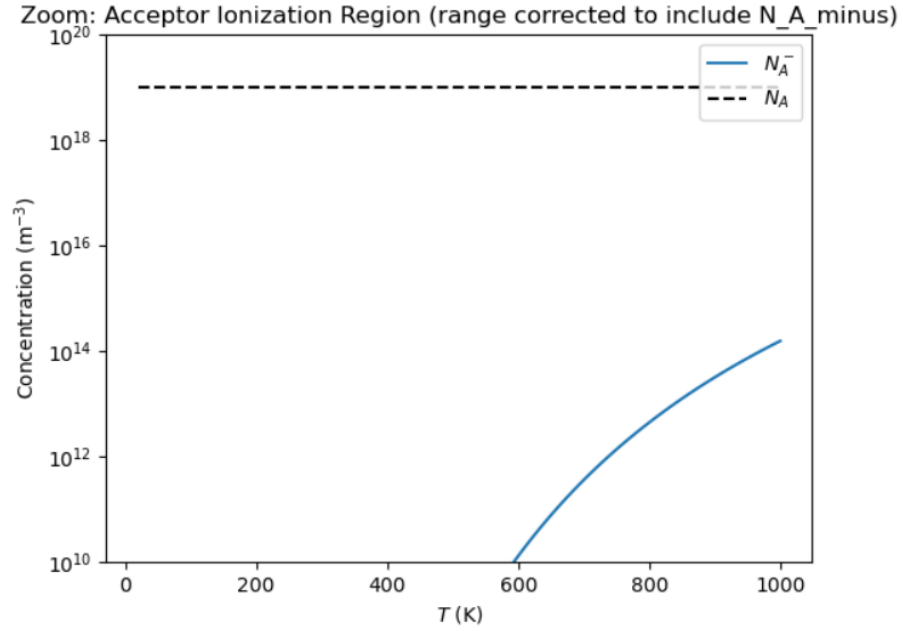


Figure 4.4: Adjusted zoom showing the actual range of the ionised acceptor concentration $N_A^-(T)$, revealing its slow increase with temperature.

4.1.3 Carrier Concentrations

The electron and hole concentrations were computed using the expressions above and plotted both on a logarithmic scale and using their natural logarithms as a function of $1000/T$, as required by the instructions.

Figure 4.5 displays the concentrations on a logarithmic scale. Due to the dominant donor density and the resulting Fermi-level position, both $n(T)$ and $p(T)$ remain extremely small throughout the temperature range.

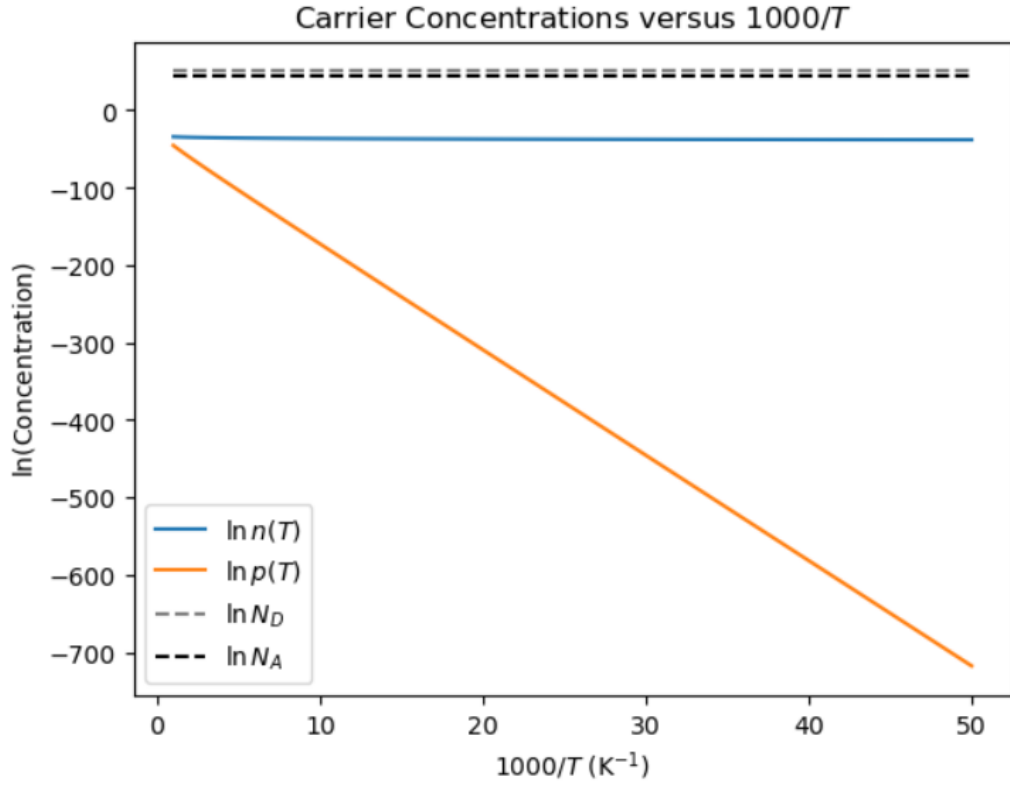


Figure 4.5: Carrier concentrations $n(T)$ and $p(T)$ on a logarithmic scale as a function of $1000/T$. The doping concentrations N_D and N_A are included as reference lines.

To better visualise the temperature dependence, Fig. 4.6 plots $\ln(n)$ and $\ln(p)$ versus $1000/T$. The nearly constant $\ln(n)$ and the linear decrease of $\ln(p)$ reflect the expected activation behaviour under strong donor doping.

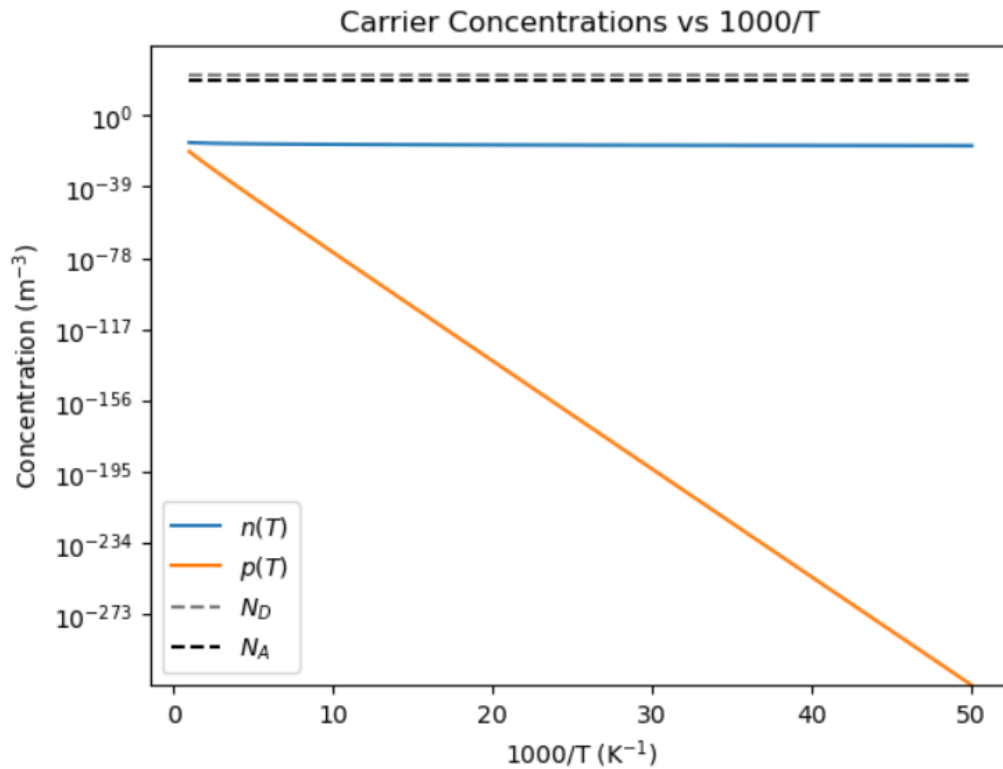


Figure 4.6: Natural logarithm of electron and hole concentrations as a function of $1000/T$. This representation highlights the activation behaviour of carriers at higher temperatures.

4.2 Silicon Sample

In this section, the electrical transport properties of the silicon sample are analysed as a function of temperature using Hall-effect measurements. From the measured voltages and resistances, the charge carrier density, electrical conductivity, Hall mobility, and an estimate of the Fermi level are obtained. The results are discussed following the tasks given in the lab manual.

4.2.1 Temperature-dependent properties

Figure 4.7 shows the charge carrier density plotted as a function of inverse temperature ($1000/T$) on a semi-logarithmic scale. The carrier density increases continuously with temperature. This behaviour indicates thermally activated charge carriers, which is expected for doped silicon.

At lower temperatures, the data follows an approximately linear behaviour in the semi-logarithmic plot. This region can be described by an Arrhenius law and corresponds to incomplete ionization of donor atoms. From this fit, an effective donor ionization energy is extracted. At higher

temperatures, the measured values deviate from the Arrhenius fit, indicating the transition into the extrinsic regime, where most donors are ionized.

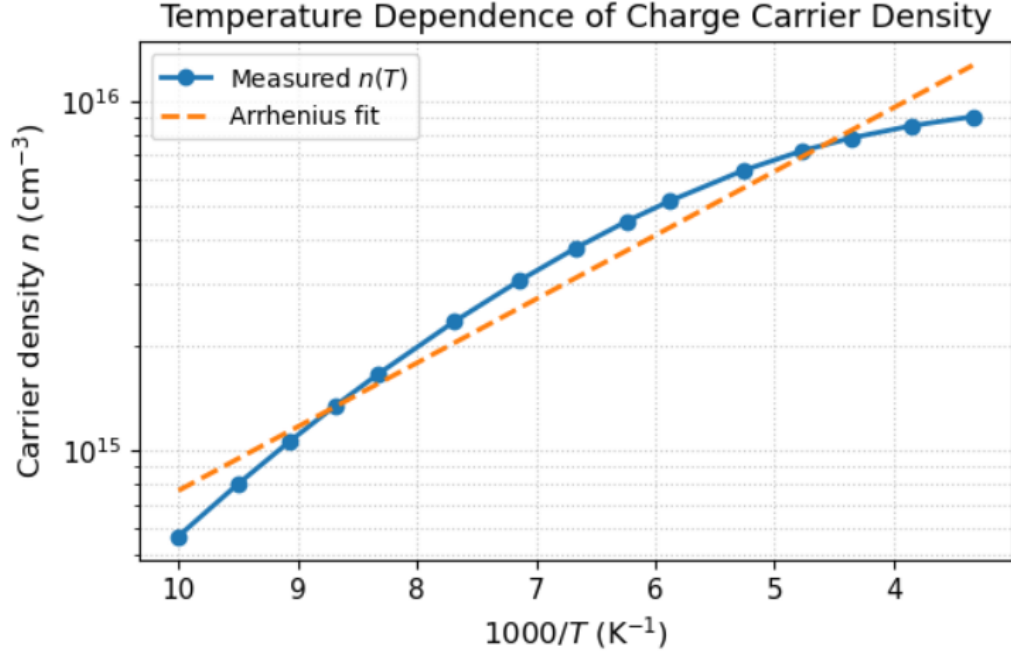


Figure 4.7: Temperature dependence of the charge carrier density $n(T)$ plotted versus $1000/T$. The dashed line shows an Arrhenius fit used to estimate the donor ionization energy.

The electrical conductivity as a function of inverse temperature is shown in Fig. 4.8. The conductivity increases with temperature at low temperatures due to the increasing carrier density. At higher temperatures, the conductivity decreases again. This reduction is caused by the decreasing mobility of charge carriers due to enhanced phonon scattering. The observed maximum in conductivity therefore results from the competing effects of increasing carrier density and decreasing mobility.

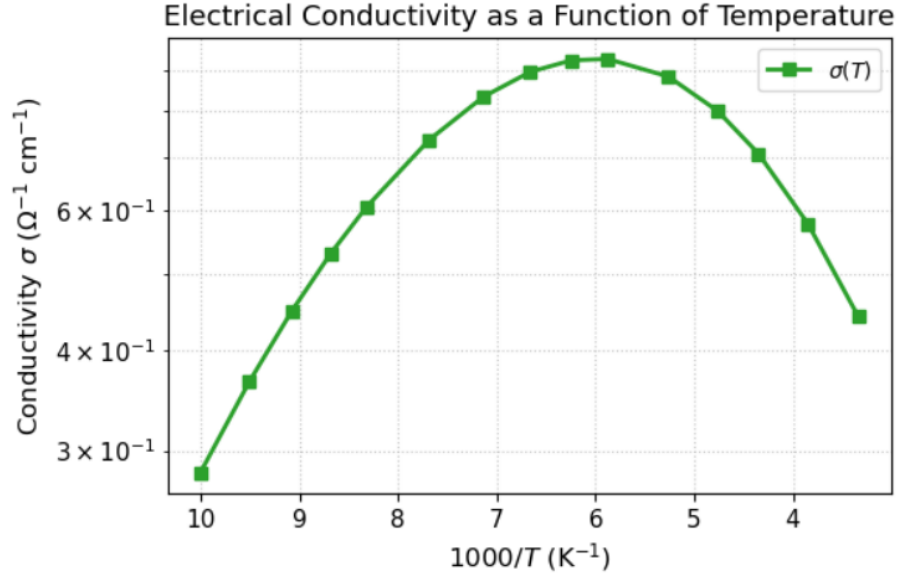


Figure 4.8: Electrical conductivity $\sigma(T)$ plotted versus $1000/T$. The non-monotonic behaviour results from the interplay between carrier density and mobility.

Figure 4.9 shows the Hall mobility as a function of temperature on a double-logarithmic scale. The mobility decreases with increasing temperature and follows a power-law dependence of the form $\mu_H \propto T^x$. From a linear fit, an exponent of $x \approx -2.1$ is obtained. This behaviour is characteristic of phonon-dominated scattering, which is expected for silicon in this temperature range.

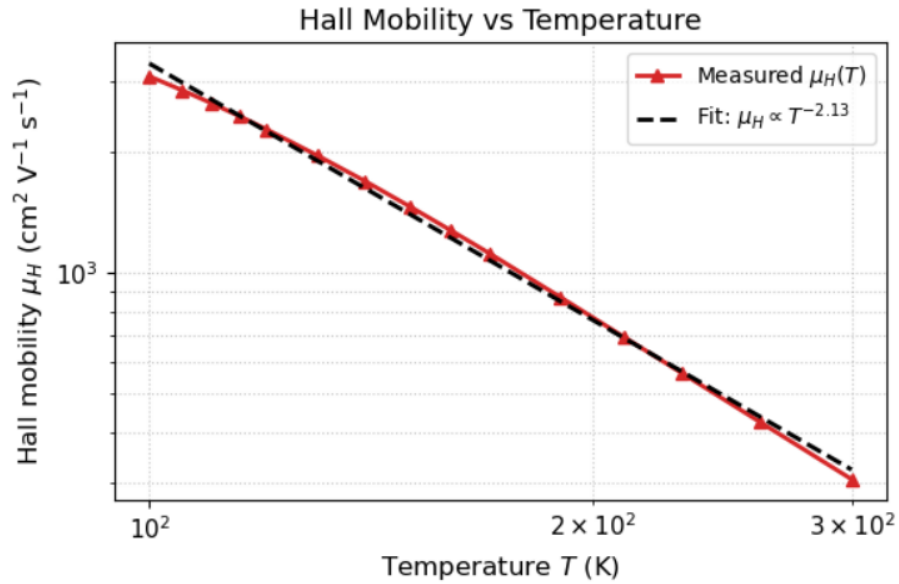


Figure 4.9: Hall mobility $\mu_H(T)$ as a function of temperature. The dashed line represents a power-law fit indicating phonon-limited charge transport.

4.2.2 Determination of sample parameters

The sign of the Hall coefficient is negative over the entire temperature range. This indicates that electrons are the majority charge carriers and that the silicon sample is **n-type**.

From the Arrhenius analysis of the carrier density, a donor ionization energy of approximately

$$\Delta E_D \approx 36 \text{ meV}$$

is obtained. According to literature values from the IOFFE Institute database, the donor ionization energy for phosphorus-doped silicon is approximately 45 meV [6]. The slightly lower value obtained in this experiment can be explained by compensation effects, experimental uncertainties, and the simplified analysis used.

The carrier density lies in the range of 10^{15} to 10^{16} cm^{-3} , which corresponds to moderately doped silicon. The mobility exponent confirms that lattice (phonon) scattering is the dominant scattering mechanism in the investigated temperature range, which is consistent with literature data [5].

4.2.3 Estimate of the Fermi level

Using the measured charge carrier density, the temperature dependence of the Fermi level is estimated assuming non-degenerate statistics. The resulting Fermi level is shown in Fig. 4.10. With increasing temperature, the Fermi level moves towards the centre of the band gap. This behaviour is expected for n-type silicon, as thermal excitation increases the intrinsic carrier contribution and reduces the relative influence of the donor states.

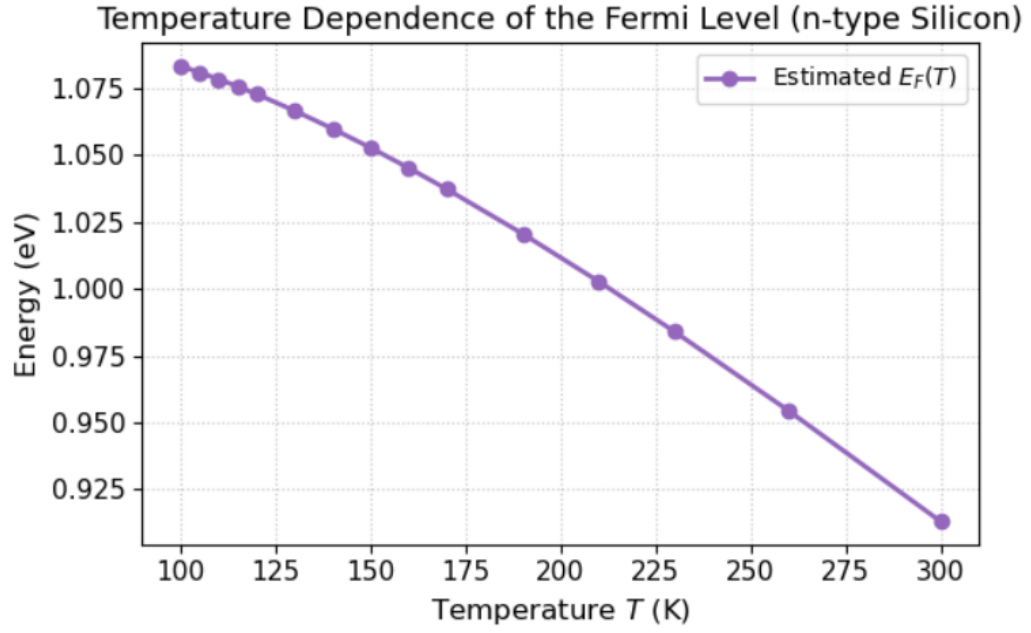


Figure 4.10: Estimated temperature dependence of the Fermi level $E_F(T)$ for the n-type silicon sample.

4.3 Indium Antimonide Sample

In this section, the temperature-dependent transport properties of an InSb sample are analysed using Hall-effect measurements. The goal is to understand the charge transport mechanisms, determine key material parameters, and identify the temperature range where the single-band Hall approximation is no longer valid.

4.3.1 Temperature-dependent transport properties

Figure 4.11 shows the semi-logarithmic plot of the absolute Hall coefficient $|R_H|$ as a function of $1000/T$. At low temperatures, $|R_H|$ decreases gradually with increasing temperature, indicating electron-dominated transport. Around $1000/T \approx 8 \text{ K}^{-1}$ (corresponding to $T \approx 125 \text{ K}$), a pronounced minimum is observed. This behaviour signals a transition where both electrons and holes contribute significantly to charge transport.

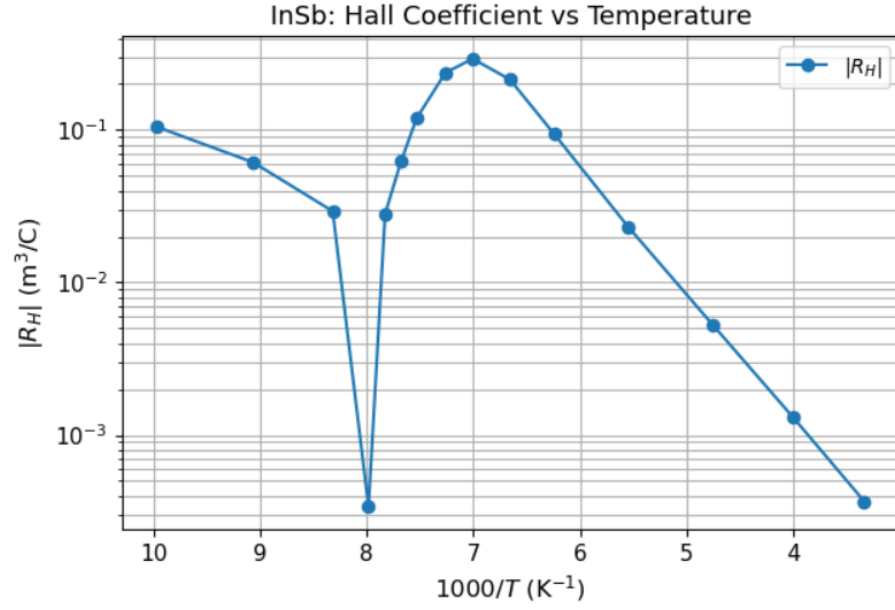


Figure 4.11: Hall coefficient $|R_H|$ of InSb as a function of $1000/T$.

The corresponding single-band Hall carrier density is shown in Fig. 4.12. At low temperatures, the carrier density remains relatively small, consistent with extrinsic conduction dominated by dopants. Near $T \approx 125$ K, an unphysical spike appears in the single-band carrier density. This is a clear indication that the single-band Hall approximation is no longer valid in this temperature range.

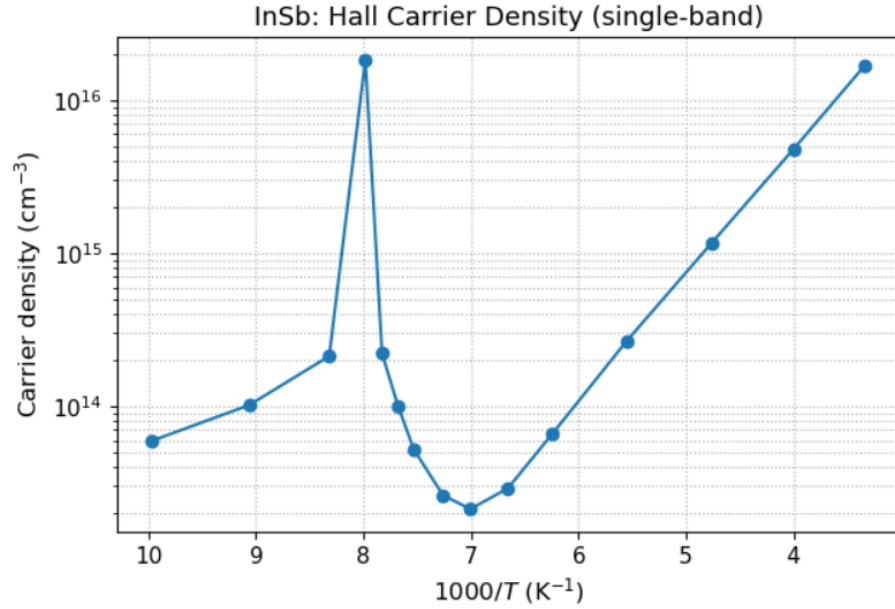


Figure 4.12: Single-band Hall carrier density of InSb versus $1000/T$.

Figure 4.13 presents the electrical conductivity σ as a function of $1000/T$. The conductivity increases strongly with temperature, which is expected for InSb due to its narrow bandgap. At higher temperatures, thermally generated charge carriers dominate the transport, leading to a rapid increase in conductivity.

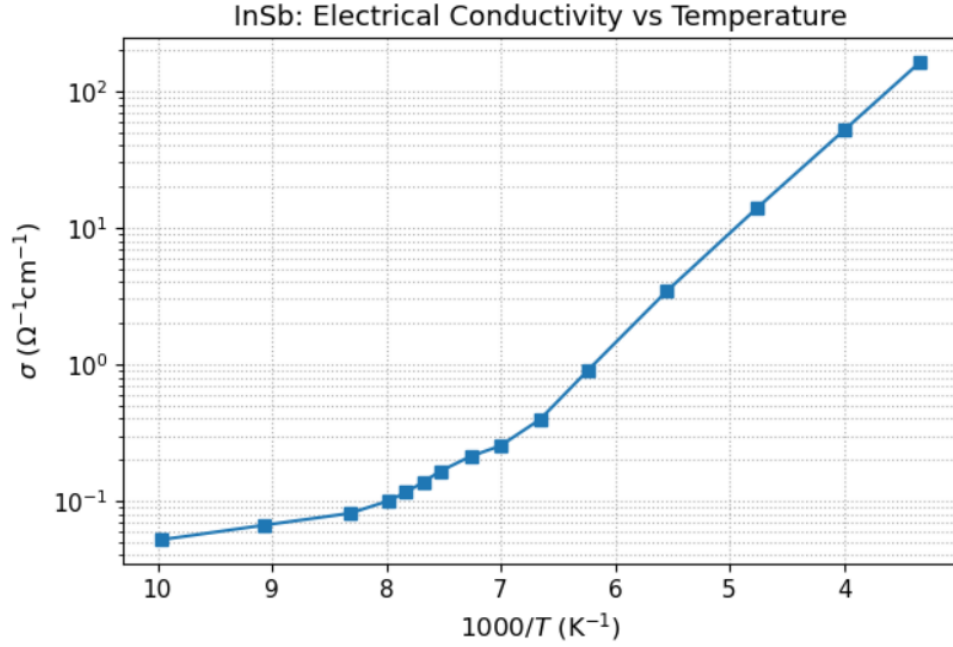


Figure 4.13: Electrical conductivity σ of InSb as a function of $1000/T$.

The Hall mobility μ_H is shown in Fig. 4.14 on a double-logarithmic scale. At low temperatures, the mobility decreases due to increased impurity and defect scattering. Above approximately 125 K, the mobility reaches very high values, exceeding 10^4 cm² V⁻¹ s⁻¹, which is characteristic of InSb and reflects its small effective mass and weak phonon scattering.

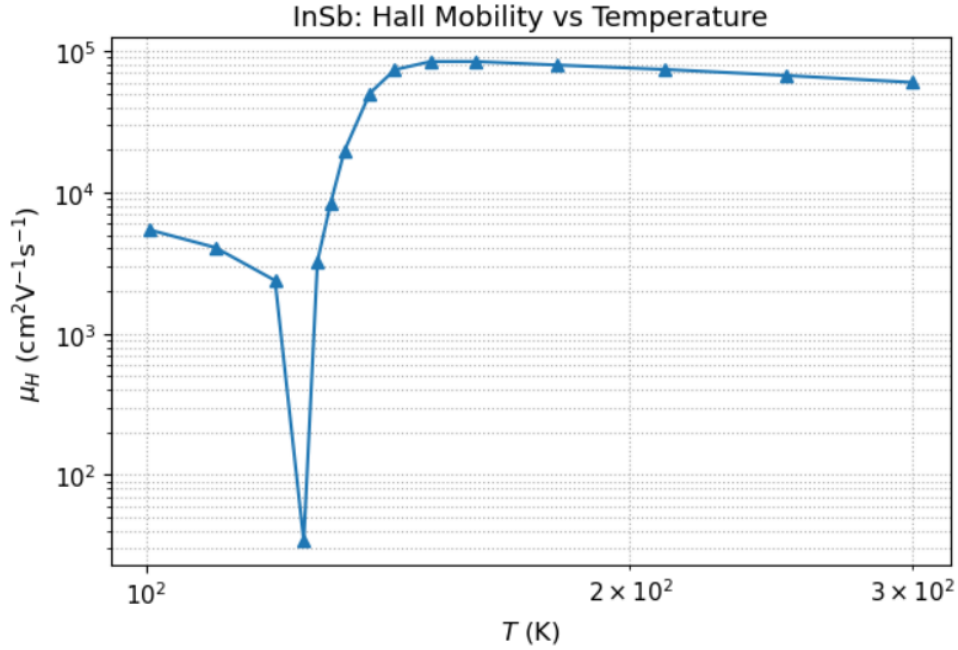


Figure 4.14: Hall mobility μ_H of InSb as a function of temperature.

4.3.2 Validity of the single-band Hall approximation

The sharp minimum in $|R_H|$ and the corresponding divergence in the single-band carrier density around $T \approx 125$ K clearly show that the assumption of a single charge carrier species breaks down in this temperature range. Above this temperature, both electrons and holes contribute significantly to conduction, and a two-band Hall model is required for a correct interpretation of the data.

4.3.3 Sample parameters

From the sign of the Hall coefficient, the conduction type of the InSb sample is identified as **n-type** over the full temperature range.

The concentration of electrically active dopants is estimated from the low-temperature carrier density, where extrinsic conduction dominates. This yields an active donor concentration of approximately

$$N_D \approx 3.8 \times 10^{15} \text{ cm}^{-3}.$$

An effective bandgap energy is extracted from the high-temperature region of the carrier density using an Arrhenius-type analysis. The resulting value is

$$E_g \approx 0.33 \text{ eV}.$$

4.3.4 Comparison with literature values

According to the IOFFE database, the bandgap of InSb at room temperature is approximately $E_g \approx 0.17$ eV. The higher value obtained in this work can be explained by several factors: the temperature dependence of the bandgap, the temperature dependence of the effective density of states in the conduction and valence bands, and the increasing importance of two-band transport effects at elevated temperatures. These effects lead to an effective bandgap value that appears larger when analysed using simplified single-band models.

4.3.5 Two-band Hall effect

The observed temperature dependence of the Hall coefficient is consistent with the two-band Hall effect in narrow-gap semiconductors such as InSb. At low temperatures, electrons dominate the transport. As the temperature increases, thermally excited holes contribute increasingly, reducing the magnitude of the Hall coefficient and causing the observed minimum. At higher temperatures, intrinsic conduction becomes dominant, and the Hall coefficient decreases further due to the combined contribution of both carrier types.

Overall, the experimental results and their temperature dependence are in good qualitative agreement with the expected behaviour of InSb reported in the literature.

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